Monitoring Outfall Pollution in Birmingham: Second Community Update

Welcome to the second update on our project assessing “surface water” outfall pollution across the Cole (Hall Green-Sparkhill) and Sutton Park.
As described in our previous roundup, this a BISCA Catalyst Fund project being led by the University and will feed into the BLOSSOM project that is looking to help address this pollution issue in east Birmingham.

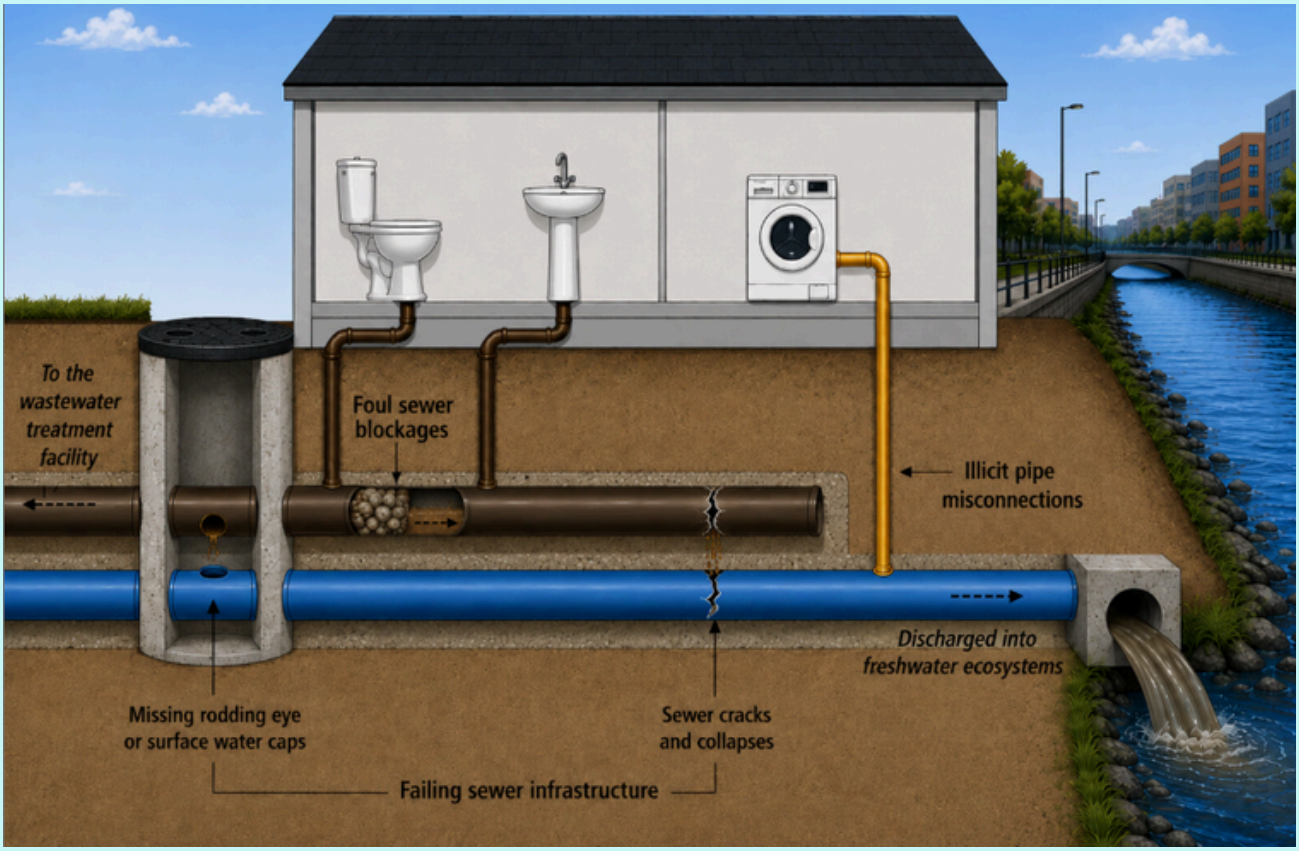
A Quick Recap:

- Some sewers transport rainwater that has entered drains directly into rivers;
- This can cause problems with things like road salts and oils;
- These issues are significant worsened by sewage contamination.

This update provides details on surface water outfalls across the areas being studied. We present results from Outfall Safaris, and show where these are located. We also share some new insights and observations from field surveys.

Illustrating Contamination

The image below shows how outfall pollution can occur – this is an exclusive from our upcoming publication that we’ll send out once its online!



What Is ‘Urban Outfall Safari’?

Urban Outfall Safari allows volunteers to capture if, and to what extent, outfalls are polluting river environments. It provides a rapid assessment using an simple survey (only three question) that can be performed from the riverbank. The three questions (which surveyors answer by providing a score on a scale of 1-5) are: How much water is there?; What’s the severity of the aesthetics (smell, appearance); How far do these impacts extend downstream?

Form more information on Urban Outfall Safari visit the Birmingham River Champions website.

Monitoring Outfall Pollution Along The River Cole

Stratford Road Bridge Outfalls

Pollution from outfalls at Blackberry Way was first observed by Birmingham River Champions (BRC) in 2024. This has been flagged to Severn Trent several times since, who always send an engineer to inspect/take ammonia readings. Their internal notes include:

“Took sample reading which was 1.88 at the outfall…lifted upstream storm drain…with reading of 0.32, no ammonia smell…” [the measurements are presumed as total ammonia in mg/L]

“…took a sample from outfall reading as slight signs of ammonia spoke to senior this is an ongoing misconnection seniors are aware of and are chasing it up”

While it's great to see these investigations taking place, how they lead to action is not clear. A Severn Trent employee has indicated that a major misconnection had been resolved, but the issue persists, suggesting other contamination sources.



Water Chemistry Sampling

Monitoring at Stratford Road bridge recently revealed a **highly acidic sample (pH = 1.6)** from the left bank outfall. While only observed once, it’s worrying to see what chemicals go into our rivers!

We still plan to install **high-frequency sensors** here that record pollution levels every 15-minutes. However, our recent installation efforts were thwarted by the concrete-lined riverbed (the smiles didn’t last!), likely dating to the bridges’ construction in 1914. We’ll return next month with more robust equipment.

New Pipe Discharges





Volunteers and university staff have observed nasty pollution being discharged out of small pipes at Blackberry Way (coming out of the shops just downstream of the Stratford Road bridge outfalls- **///smiled.next.career**). While not constantly polluting, a brown liquid has been observed. These are private drains and not Severn Trent's legal jurisdiction, but the issue has been flagged with the Environment Agency and the Council.

River Cole Outfall Safaris

University staff recently collected Outfall Safaris at all observable surface water outfalls along the Cole between The Dingles and Blackberry Way (guided by sewer maps provided by Severn Trent's River Rangers). The results are presented below, where the score is based on: A = 0; B = 2; C = 4; D =6 and E = 10. Scores over 6 indicate a polluting outfall, while those over 10 indicate significant pollution.

While those at the Stratford Road bridge were the only 'polluting' outfalls detected, intermittent pollution can be missed. Repeating Outfall Safari surveys increases the chances of spotting pollution that can arise at any time! We would therefore welcome any Outfall Safari entries via the *Birmingham River Champions* website, but please ensure that you are registered with the project.



River Cole Outfall Safaris
W3W Location: ///wings.tilt.begin
Flow: **D** Distance: **D** Aesthetics: **D**
Score (dist + aesth): 12

As indicated above, this is a persistent polluter. This is located on the left bank (looking downstream) – underneath the shops. 'Sewage fungus' (actually bacteria) is consistently evident downstream. Viewing the outfall is slightly tricky. It can (just) be seen from the bridge, but we wouldn't recommend leaning over. The easiest option would be to access the river channel from the concrete ramp in Blackberry Way and walk to the opposite bank towards the bridge.

Name: Blackberry Way (right)
W3W Location: ///legal.fled.influencing
Flow: **D** Distance: **D** Aesthetics: **C**
Score (dist + aesth): 10



This also consistently pollutes, and while the 'aesthetics' of the water itself do not rank quite as highly as the left bank, toilet paper is always present. Pollution from this outfall was first observed a little later (June 2025) by university staff.

As above, the easiest way to view this is from in the channel accessible via the concrete ramp.





Name: Green Road ford (left and right bank)
W3W Location**: ///**hers.this.truly **and**
///jams.going.vision
Flow: **A** Distance: **A** Aesthetics: **A**
Score (dist + aesth): 0 (same for both)

No discharge was observed at either outfall. The left bank outfall is under dense vegetation, but water discolouration from pollution should be visible from the footbridge. The outfall on the right bank can be observed from the ford. There is a safe access point into the Cole 30-metres downstream on the left bank, where you can walk back upstream to see these.

Name: Sarehole Mill millstream
W3W Location: ///lasted.paper.detail
Flow: **B** Distance: **B** Aesthetics: **B**
Score (dist + aesth): 4



Water was present in the outfall, but flow was minimal. Some foam was spotted potentially indicating pollution, but this was not severe and very localised. The site can be easily observed from over the railings next to the footpath. There is also an access point across the millstream just to the right of the photo (towards the ford) which would provide a better view of the outfall.





Name: Cole Bank Road bridge (left and right)
W3W Location: ///shin.digs.next **and** ///suffer.easy.hers
LEFT Flow: **B** Distance: **B** Aesthetics: **B Score (dist + aesth): 4**

RIGHT Flow: **B** Distance: **A** Aesthetics: **A** Score (dist + aesth): **0**
Both outfalls under the bridge. The left bank (left) is tucked further away. There seemed to be some discharge and 'grey-ish' water, but was not clear due to the darkness. No discharge evident in the right bank, but much more visible from the left bank. There is a safe access point into the channel from Sarehole Mill car park.

Name: John Morris Way (north)
W3W Location: ///rather.sunk.discrepancy
Flow: **B** Distance: **B** Aesthetics: **B**
Score (dist + aesth): 4

There was a minimal amount of water coming out but there was a small area of 'grey-ish' water beneath - but unclear if this was pollution. Located on the right bank (opposite to the footpath) and is best visible from the channel, where there are access points nearby from the left bank.





Name: John Morris Way (south)
W3W Location: ///strict.onions.woods
Flow: **A** Distance: **A** Aesthetics: **A**
Score (dist + aesth): 0

No water seen being discharged. As above, located on the right bank but is more easily visible from the left bank next to footpath and there are easier access points here.

Name: Coleside Avenue
W3W Location: ///bonds.hope.cafe
Flow: **A** Distance: **A** Aesthetics: **A**
Score (dist + aesth): 0



A smaller pipe than most other surface water outfalls. If you follow a straight line from the rivet to the houses on Coleside Avenue, it's in line with the final terraced house. You can see this by stepping over the small fence and looking over the bank face. There is a large drop down to the channel from the bank, and access to the channel is not advised.



Name: The Dingles (north)
W3W Location: ///coins.areas.human
Flow: **A** Distance: **A** Aesthetics: **A**
Score (dist + aesth): 0

No water or pollution being discharged. A large outfall that is located close to the footpath just before the Cole flows out of The Dingles. The best view is from the channel (as photographed), although pollution could be seen by peering over from the riverbank (but there is a drop).

Name:The Dingles (confluence)
W3W Location: ///race.soon.risen
Flow: **A** Distance: **A** Aesthetics: **A**
Score (dist + aesth): 0



No water seen being discharged. This is located directly where the Chinn Brook joins the Cole. Its very easy to see, but requires walking a few metres off the footpath (cross the last small footbridge over the Chinn and turn left), and care should be taken here.



Name:The Dingles (the gap)
W3W Location: ///kick.ends.will
Flow: **A** Distance: **A** Aesthetics: **A**
Score (dist + aesth): 0

No water seen being discharged. Situated on the right bank (Yardley Wood side), it’s challenging to view due to vegetation on the left bank. This can just about be seen once you are able to reach the top of the left bank.

Monitoring Outfall Pollution Across Sutton Park

River Cole Outfall Safaris

University staff have collected various Outfall Safaris in recent months across Sutton Park. The outfalls assessed have been identified by Severn Trent river rangers who have kindly supported this process.

All outfalls have consistently been found not to be discharging water or any kind of pollution (but we did notice once exception to this recently - see below).

As highlighted above, repeat Outfall Safari surveys as part of BRC can help capture intermittent pollution and new contamination.

Water Chemistry

In our previous community update, we presented water chemistry findings at one outfall in Sutton Park. We should clarify this particular outfall contains inputs from both Combined Sewer Overflows (CSO) and (separate) surface water sewers. This surface water outfall is the only one where we've spotted pollution during recent visits. We have continued collecting water chemistry and will draw on different evidence to investigate any potential pollution sources, but nothing unexpected has been measured so far.





Name: Longmoor trib joint CSO
W3W Location: ///bonus.pitch.bronze
Flow: **B** Distance: **B** Aesthetics: **B**
Score (dist + aesth): 4

The surface water outfall is marked with a star (the other outfalls are from CSOs, but also contain groundwater). On one occasion, we observed a foamy substance being discharged from here (less likely to be sewage), producing the scores above. But in all other site visits no water or pollution has been observable

Name: Longmoor trib Chester Road 1&2
W3W Location:///cloak.scans.coffee **and** ///luxury.crab.settle
1 & 2. Flow: **A** Distance: **A** Aesthetics: **A**
Score (dist + aesth): 0 (same for both).





No water being discharged observed at either outfall. Both are situated along a Longmoor Stream tributary draining from the Chester Road. The outfall 1 is positioned on the main channel itself, while 2 is located just off from this nearby. Both are easily viewable from footpaths immediately south of Lord Donegal's ride.



Name: Ebrook trib Blackroot Pool
W3W Location: ///riots.cans.finishing
Flow: **A** Distance: **A** Aesthetics: **B**
Score (dist + aesth): 2

No discharging water or pollution has been observed from the outfall. A faint smell was observed here on one occasion. The outfall is easily viewable from where the footpaths join close to the railway line.

Name: EBrook trib Meadow Platt
W3W Location: ///blows.famed.rainy
Flow: **A** Distance: **A** Aesthetics: **A**
Score (dist + aesth): 0

No discharging water or pollution has been observed from the surface water outfall. The best access does requireclimbing through some vegetation and and walk down the river.





Name: EBrook trib Leisure Centre
W3W Location: ///wounds.slam.galaxy
Flow: **A** Distance: **A** Aesthetics: **A**
Score (dist + aesth): 0

No pollution or water has been observed, but the outfall is very conspicuous and we are reaching out to Severn Trent for further confirmation of this outfall. It is located just downstream of the footbridge along the footpath close to the playground / leisure centre.

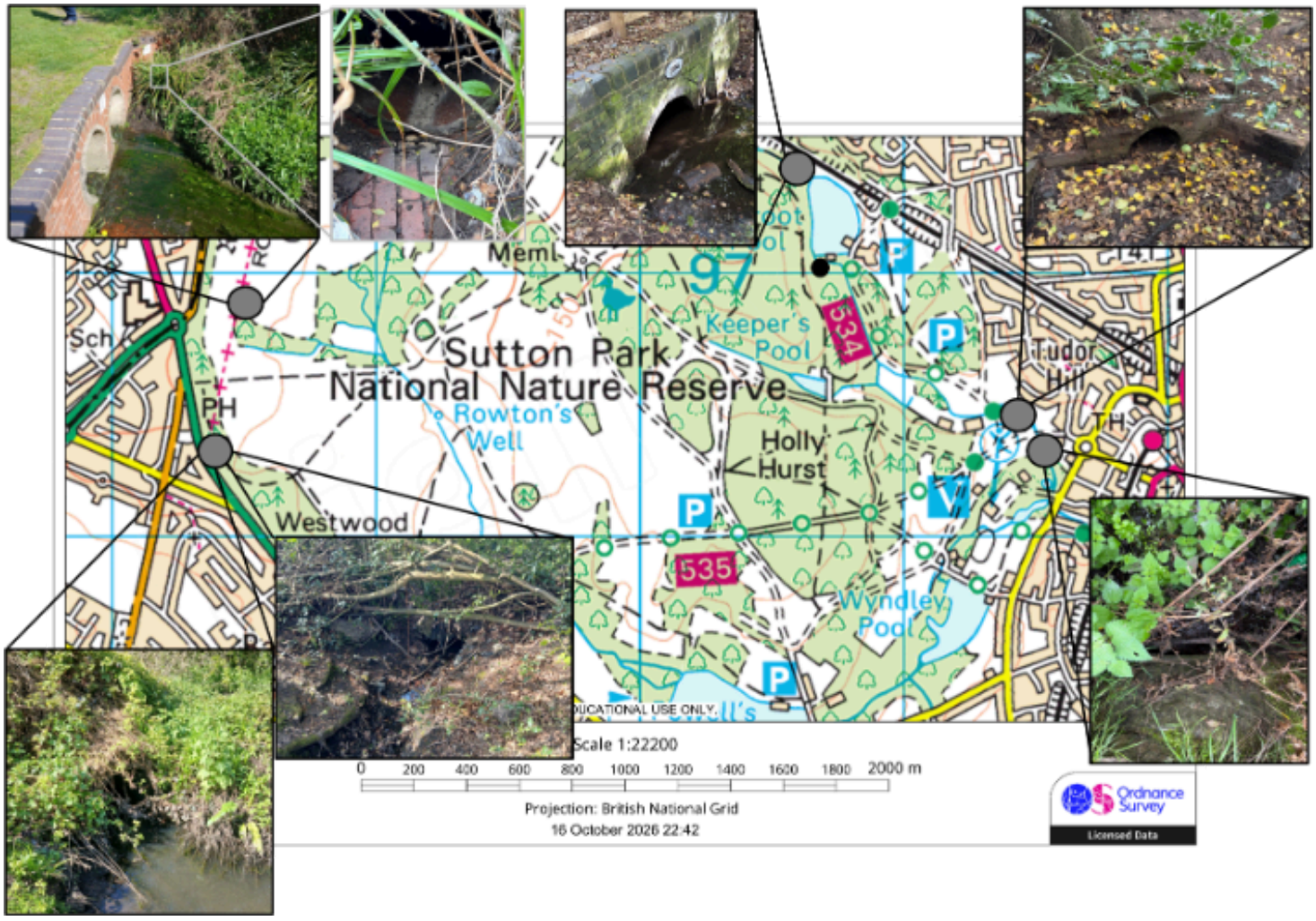
Summary

Overall, the outfall safaris have indicated that visible pollution threats from surface water outfalls are low across Sutton Park. While along the Cole ‘highly polluting’ outfalls have been recorded at the Stratford Road Bridge in Sparkhill, and while not clear polluting some outfalls warrant more regular inspections (‘John Morris Way (North)’ and ‘Cole Bank Road bridge (left)’). Any community support in collecting this data via the Birmingham River Champions would be gratefully received and will be shared with Severn Trent. Overleaf you can see maps of these outfalls described.

A map of surface water outfalls along the Cole: Outfall Safaris and location descriptions are provided above. Some surface water outfalls may have been present that were not visible. Other outfalls may also be evident (e.g., see top right), but these could represent private drains or CSO outfalls, which are different.



A map of surface water outfalls across Sutton Park: Outfall Safaris and location descriptions are provided above. Some surface water outfalls may have been present that were not visible. Other outfalls may also be evident, but these could represent private drains or CSO outfalls, which are different.



Birmingham River Champions

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